

Introduction to Machine Learning

DIGITAL LINGUISTICS AND
LANGUAGE PROCESSING

Structure

- Fundamental concepts in Machine Learning
 - Data
 - Algorithms
 - Experimental design
 - Evaluation
- Short demo & Hands-on exercise
 - Designed for complete beginners

What is Machine Learning?

- **Learn from data:** Machine learning is a branch of artificial intelligence that focuses on building systems that can *learn from and make decisions based on data*.
- **No pre-defined rules:** Unlike traditional programming, where explicit instructions are given, machine learning models identify patterns and make decisions with *minimal human intervention*.

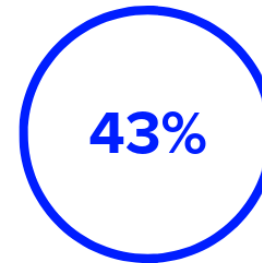
Why Machine Learning?

- **Adaptability:** ML systems can adapt to new data independently, allowing them to evolve and improve over time as they are exposed to more data.
- **Automation:** ML can automate decision-making processes in various applications (e.g., speech recognition, AI chatbots, resume screening).
- **Innovation and Efficiency:** ML drives innovation in many fields, such as healthcare (e.g., drug discovery, medical imaging).
- **Pervasiveness:** ML has become integral to many services and technologies that enhance our daily lives (for the majority of the time, we aren't even conscious of it.)

Role of Data in Machine Learning

- Role of Data in Machine Learning
 - Importance of high-quality data
 - „garbage in, garbage out“
- Data collection
 - Methods of data collection (manual, automated)
 - Examples of data sources (web scraping, surveys)
- Types of data
 - Structured data: tables, databases (e.g., SQL)
 - Unstructured data: text, images, audio
- Data preparation
 - Data cleaning, normalization, transformation
 - Handling missing values and outliers

Data quality is biggest ML challenge



say poor data quality
is biggest ML barrier



report lack of data
availability as 2nd
highest barrier



find it hard to hire
data-scientist talent

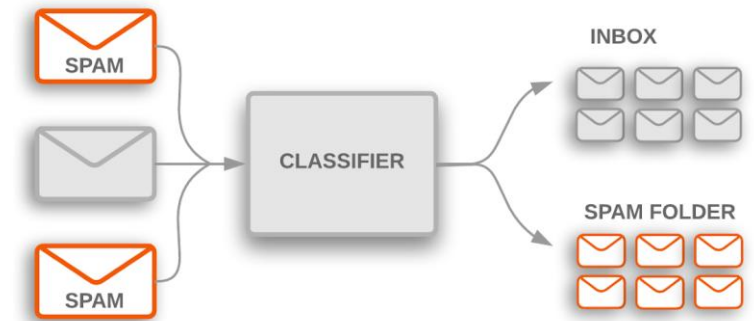
#MLReadyData

Source: Refinitiv AI/ML Survey

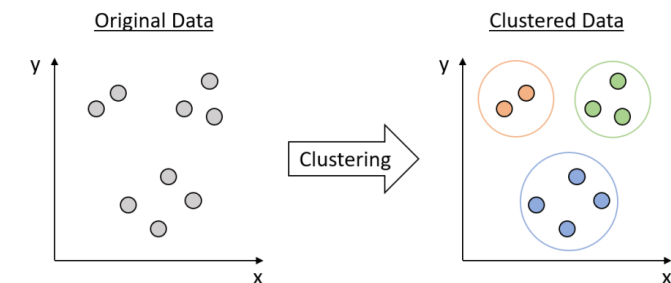
REFINITIV™
DATA IS JUST
THE BEGINNING

Learning settings

- Supervised Learning
 - A type of machine learning where the model is trained on **labeled data**.
 - The algorithm learns **to map input data to the correct output** based on provided examples.
- Unsupervised learning (clustering)
 - A type of machine learning where the model is trained on **data without labels**.
 - The algorithm tries to find hidden patterns or intrinsic structures in the input data.



Source: <https://cloud.google.com/blog/topics/developers-practitioners/how-spam-detection-taught-us-better-tech-support>



Source: <https://waterprogramming.wordpress.com/2022/03/16/clustering-basics-and-a-demonstration-in-clustering-infrastructure-pathways/>

Machine learning tasks

- **Regression**
 - A type of supervised learning used to predict continuous outcomes based on input features.
- **Classification**
 - A supervised learning method used to categorize data into predefined classes or labels.
- **Clustering**
 - An unsupervised learning technique used to group similar data points into clusters based on inherent patterns.

Basic Algorithms in Machine learning

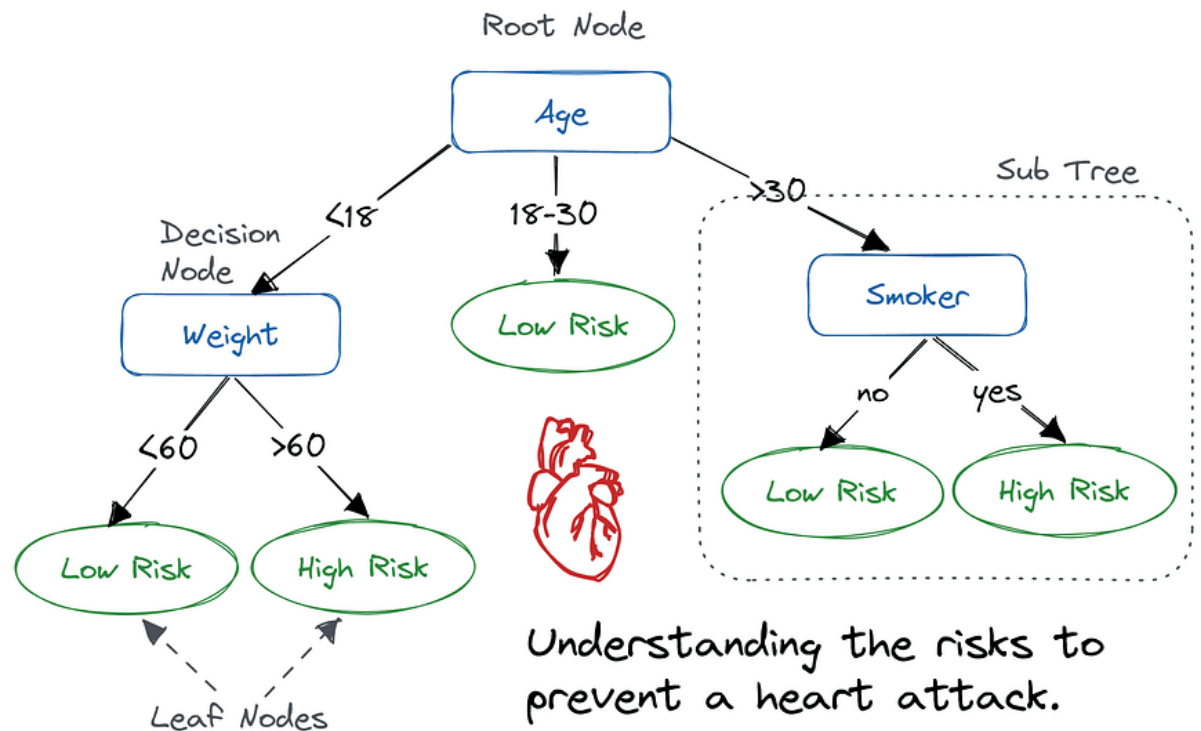
Dummy classifier or regressor

- It's purpose is to establish a baselin
- Classification -> Majority classifier
- Regression -> Take the mean of the values
- Usually our first goal is to improve over this baseline

Basic Algorithms in Machine learning

Decision trees

- A decision tree is a supervised learning algorithm used for both classification and regression tasks.
- It models decisions and their possible consequences as a tree structure of nodes and branches.
- **How it works?**

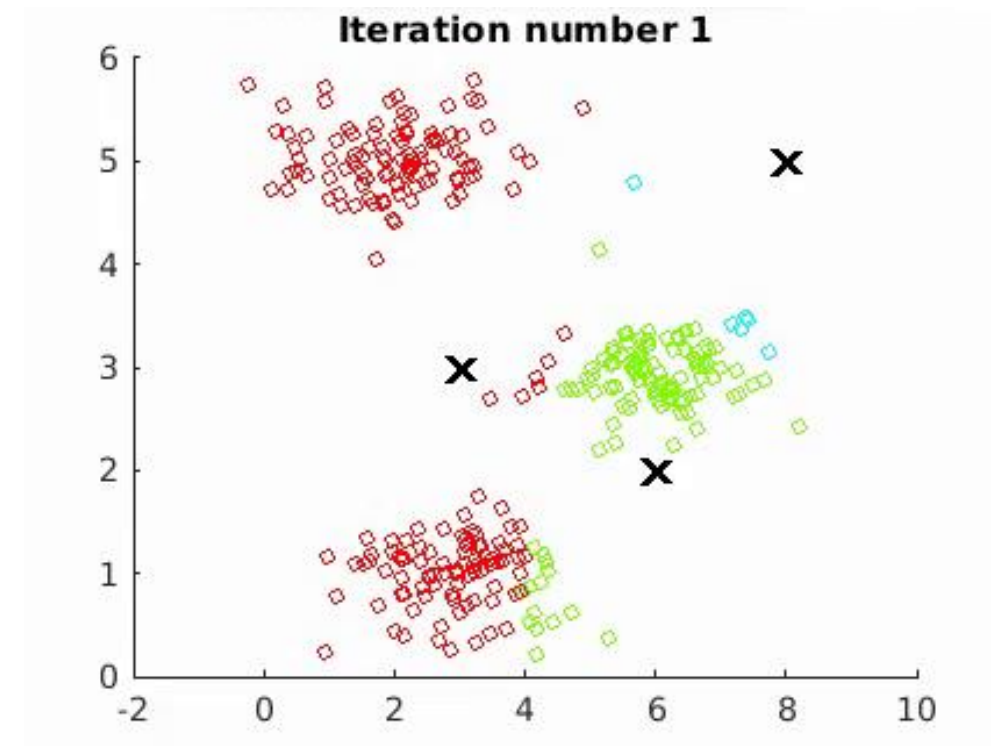


Source: <https://www.datacamp.com/tutorial/decision-tree-classification-python>

Basic Algorithms in Machine learning

K-means clustering

1. Initialization:
 - Select the number of clusters, k
 - Randomly initialize k centroids (center points of clusters).
2. Assignment Step:
 - Assign each data point to the nearest centroid.
 - This forms k clusters based on proximity.
3. Update Step:
 - Calculate the new centroid of each cluster by computing the mean of all data points assigned to that cluster.
 - Update the centroid positions to these new values.
4. Repeat:
 - Repeat the assignment and update steps iteratively until:
 - The centroids no longer change significantly (convergence).
 - The assignments of data points to clusters do not change.



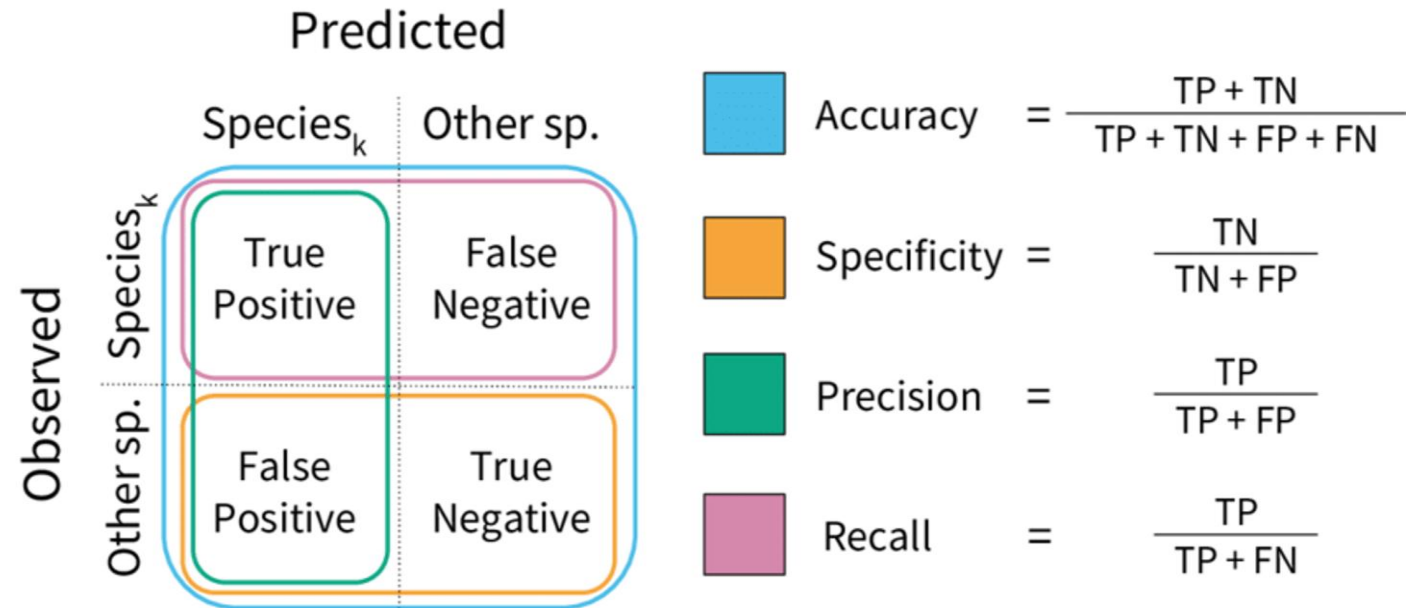
Source: <https://github.com/exercism/python/issues/2999>

Properties & Trade-offs of Algorithms

- **Accuracy:** The measure of how often the algorithm correctly predicts or classifies outcomes.
- **Complexity:** Refers to the intricacy of the model, including the number of features, parameters, and computational requirements.
- **Interpretability:** The ability to understand and explain how the model makes decisions or predictions.
- **Speed:** The time it takes for an algorithm to train on a dataset and make predictions.

Model evaluation

- **Accuracy** (classification)
 - Measures the proportion of correct predictions out of total predictions.
 - Note: Can be misleading in *imbalanced datasets*
- Alternatives: **F1-Score**
 - The harmonic mean of precision and recall, balancing the two



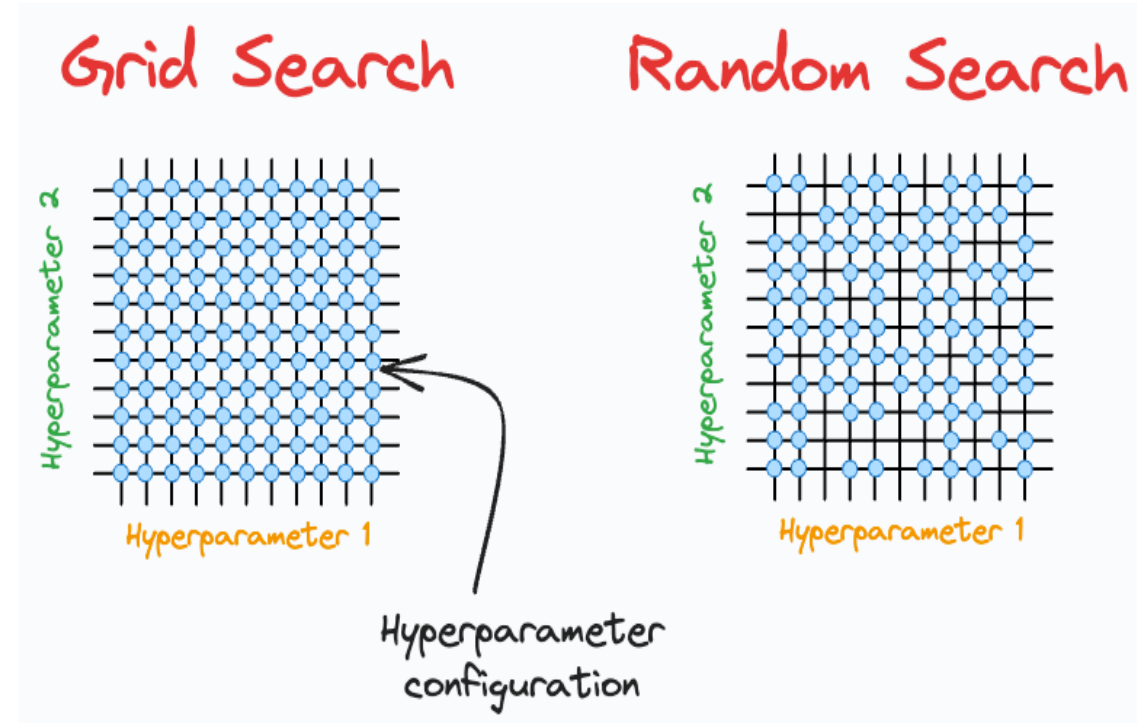
Source: <https://www.almabetter.com/bytes/tutorials/data-science/classification-metrics>

Splitting Data

- **Training Set**
 - Purpose: Used to train the machine learning model.
 - Function: The model learns the relationships within the data and develops patterns based on input features.
 - Size: Typically comprises the majority of the dataset (e.g., 60-80%).
- **Validation Set**
 - Purpose: Used to fine-tune the model's hyperparameters and prevent overfitting.
 - Function: Helps in adjusting model settings and selecting the best model architecture.
 - Size: Smaller than the training set, often 10-20% of the total data.
- **Testing Set**
 - Purpose: Used to evaluate the final performance of the trained model.
 - Function: Provides an unbiased evaluation of the model's effectiveness on new, unseen data.
 - Size: Similar in size to the validation set, usually 10-20% of the total data.
 - **Note: This set should not be used during the training or tuning process!!!**

Hyperparameter tuning

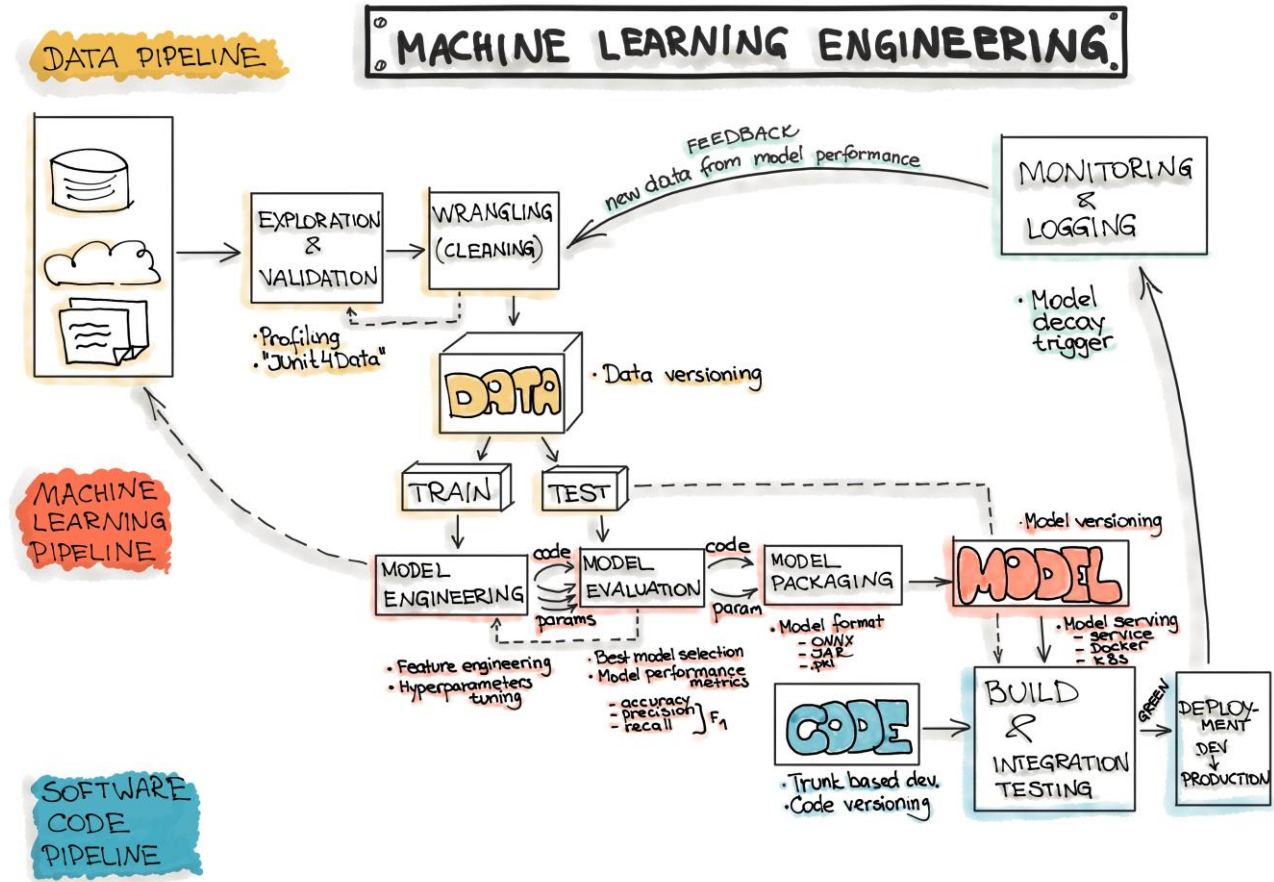
- What are **Hyperparameters**?
 - Hyperparameters are configurations of the model set *before the learning* process begins.
 - They differ from model parameters, which are learned *during training*.
 - Examples: number of trees, max depth, minimum number of samples in a leaf node
- Methods
 - Cross-validation
 - Grid search
 - Random search



Source: <https://blog.dailydoseofds.com/p/the-overlooked-limitations-of-grid>

ML process

- Data's Critical Role:
 - Essential for training robust models.
 - Proper data preparation (cleaning, normalization) is key.
- Core ML Tasks and Algorithms:
 - Regression, Classification, Clustering
 - Algorithm Selection: trade-offs in complexity and interpretability.
- Model Evaluation:
 - Metrics like accuracy, F1-score, and confusion matrices assess performance.
 - Hyperparameter tuning (e.g., grid search) optimizes models.
- Importance of Experimental Design:
 - Ensures reliable and reproducible results.
 - Techniques include train/test splits and cross-validation.



Source: <https://ml-ops.org/content/end-to-end-ml-workflow>